

RadClinNet: Multimodal Fusion of Chest CT and Clinical Variables for Survival Prediction in Non-Small Cell Lung Cancer

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Abstract

Non-small cell lung cancer (NSCLC) remains the leading cause of cancer mortality worldwide, yet clinical risk models for NSCLC survival typically rely solely on structured electronic health record (EHR) variables, while diagnostic CT scans, routinely acquired during workup, are rarely integrated into quantitative prediction. I present RadClinNet, a multimodal pipeline that fuses chest CT imaging with EHR-derived clinical features for binary survival classification on the NSCLC-Radiogenomics dataset from The Cancer Imaging Archive (TCIA). I compare three architectures: a tabular MLP using clinical features only, a CT-only model based on an ImageNet-pretrained ResNet-18 with mean-pooled axial slice embeddings, and a late-fusion model that concatenates the two modality representations. To handle the limited sample size ($n=112$ labeled patients, 48 with matched DICOM studies), I apply transfer learning, random horizontal flip augmentation, dropout, weight decay, and class-weighted binary cross-entropy loss. On the held-out test set, the tabular MLP achieves an AU-ROC of 1.000, the CT-only ResNet-18 achieves 0.889, and the fusion model achieves 0.861. Grad-CAM analysis indicates that CT attention concentrates on image periphery rather than lung anatomy, a known small-data artifact. My findings highlight the tabular-dominance regime at small dataset scales and motivate future work on cross-attention fusion and larger-cohort evaluation using NLST and TCGA-LUAD.

1. Introduction

Lung cancer remains the leading cause of cancer-related death globally. It accounts for around 1.8 million deaths a year, roughly 18% of cancer mortality overall [1]. Non-small cell lung cancer (NSCLC) is the majority subtype at about 85% of cases (adenocarcinoma, squamous cell carcinoma, and large cell carcinoma) [2]. Even with targeted

therapy and immunotherapy, five-year overall survival for NSCLC stays below 25% across stages, and falls under 10% once the disease is metastatic [3]. Individualized survival prediction at diagnosis matters for treatment intensity, adjuvant therapy decisions, trial enrollment, and end-of-life planning.

Clinical risk stratification for NSCLC still rests largely on the TNM staging system, which encodes tumor size and local invasion (T), regional lymph node involvement (N), and distant metastasis (M) into prognostic categories [4]. TNM is assembled from clinical workup, imaging review, pathology, and physical examination, and stored in the electronic health record (EHR) alongside demographics, smoking history, performance status, histology, and treatment plans. Predictive models built on these tabular EHR variables, from Cox proportional hazards regression to gradient-boosted ensembles, have shown that structured clinical data alone can stratify NSCLC patients reasonably well [5, 6]. What these models throw away, though, is a source of patient-specific information that is generated as a direct byproduct of standard care: the diagnostic chest CT scan.

Volumetric chest CT is acquired in almost every NSCLC workup, to localize the primary lesion, identify nodal involvement, and screen for metastases. A DICOM study contains hundreds of axial slices encoding tumor morphology, parenchymal background, peritumoral environment, and distant anatomic context. In practice, a radiologist distills all of that into a free-text impression and a handful of structured fields. Classical *radiomics* has tried to extract quantitative imaging features from segmented tumor regions and link them to prognosis [5, 7], but it depends on manual segmentation, hand-engineered features, and pipelines that do not generalize well across institutions. Convolutional neural networks (CNNs) pretrained on natural images and fine-tuned on medical images have since reached competitive or better performance on nodule detection, malignancy classification, and outcome prediction [8, 9].

Even so, combining CT imaging with structured EHR data for survival prediction is still relatively underexplored, especially outside large industrial labs with proprietary cohorts. Most published NSCLC survival models work on one modality at a time. When fusion is attempted, it is usually done by concatenating late-stage feature vectors, without much exploration of how the two modalities actually interact at realistic dataset sizes. A few things compound this gap. Multimodal medical datasets that pair raw CT volumes with curated clinical variables and outcome labels are scarce; public cohorts rarely go beyond a few hundred patients, and even fewer have matched imaging plus complete follow-up. CT-based deep models are also data-hungry; training a CNN from scratch on tens of patients overfits aggressively, no matter how much regularization you throw at it. And when one modality (usually the tabular features) is already enough to separate the classes, the model has no real incentive to learn from the more complex one, which produces a *modality dominance* regime where imaging contributions get quietly overwritten by clinical features [10].

This paper looks at these tensions on the publicly available **NSCLC-Radiogenomics** cohort from The Cancer Imaging Archive (TCIA), a dataset of NSCLC patients with pretreatment chest CT and an associated structured clinical CSV [11]. After filtering to patients with both a binary survival label (*Alive / Dead*) and at least one DICOM series, I get 112 labeled patients and a matched imaging subset of 48, on which a stratified 75/25 train/test split is applied. I present **RadClinNet**, a multimodal pipeline that fuses EHR-derived clinical variables with mean-pooled axial CT slice embeddings from an ImageNet-pretrained ResNet-18 backbone for binary survival classification. The three architectures I compare share the same preprocessing, splits, optimizer, and class-weighted loss: a *tabular MLP* on the one-hot-encoded, standardized clinical feature vector; a *CT-only model* that encodes each of eight sampled axial slices independently and averages the 512-dimensional embeddings before a linear classifier; and a *late-fusion model* that concatenates the CT embedding with a learned tabular embedding before a small classification head. All three are trained with class-weighted binary cross-entropy, and Grad-CAM [12] is used to inspect where the CT model attends.

I deliberately keep the experimental design conservative. Rather than reporting only the final fusion model, I trace three iterations of the CT pipeline: a custom three-layer CNN trained from scratch; the same CNN with dropout, weight decay, augmentation, and class-weighted loss; and finally a transfer-learned ResNet-18. The point is to make explicit how each design choice affects performance and overfitting on a small cohort. The pattern that emerges is well known in principle but rarely documented end-to-end at the coursework or applied-write-up level: at small medical scales, transfer learning from natural images dominates

regularization tweaks, and the choice of backbone matters far more than the choice of optimizer.

The results read as a kind of warning. On the matched test set the tabular MLP, fed only with the clinical CSV, hits an AUROC of 1.000, while the CT-only ResNet-18 reaches 0.889 and the late-fusion model lands at 0.861. The clinical CSV already encodes the post-staging information that drives short-term survival; the imaging signal, although learnable, is not additive under a concatenative fusion design. Grad-CAM overlays also show that the CT encoder, without explicit lung segmentation or anatomical priors, attends to image periphery rather than parenchymal tumor regions, a known small-data artifact in medical CNNs [13]. I read this as a characterization of what might be called the small-cohort tabular-dominance regime, and it motivates two directions for future work: attention-based fusion architectures that let imaging features modulate (rather than be concatenated with) clinical embeddings, and evaluation on substantially larger public cohorts like the National Lung Screening Trial (NLST) and TCGA-LUAD, where the tabular ceiling can no longer be saturated so easily.

Contributions. The contributions of this paper are: a reproducible CT + EHR fusion pipeline on a public TCIA cohort, with DICOM-level preprocessing (Hounsfield unit conversion, lung-window clipping, slice sampling, multi-frame handling), tabular encoding (one-hot with explicit missingness, most-frequent imputation, standardization), and stratified splitting at both the clinical-only and matched-imaging levels; a benchmark of three architectures (tabular MLP, CT-only ResNet-18, late-fusion) under identical training conditions, with an iteration path from a custom-from-scratch CNN to a transfer-learned backbone that quantifies the AUROC gain attributable to ImageNet pretraining; and a Grad-CAM interpretability analysis paired with an honest discussion of the modality-dominance regime that shows up at small scales, with concrete next steps (cross-attention fusion, larger-cohort evaluation) instead of overclaiming generalization from a small held-out test set.

The rest of the paper is organized as follows. Section 2 reviews related work on radiomics, deep learning for CT-based prognosis, and multimodal fusion in clinical risk modeling. Section 3 describes the dataset and preprocessing pipeline. Section 4 covers the three architectures and the training protocol. Section 5 reports quantitative results, Grad-CAM analysis, and the iteration ablation. Section 6 discusses limitations and future work, and Section 7 concludes.

2. Related Work

This project sits at three intersections: quantitative analysis of lung CT for prognosis, multimodal fusion of imaging

with structured clinical data, and the practical difficulties of training multimodal networks on small medical cohorts.

2.1. Radiomics and Deep Learning for Lung CT

For a long time, quantitative CT analysis for outcome prediction has been a radiomics problem. *Radiomics* extracts hundreds of hand-engineered features (intensity histograms, shape descriptors, texture filters, wavelet decompositions) from manually or semi-automatically segmented tumor regions, and feeds them into classical machine-learning models [5–7]. It established that CT-derived features carry prognostic information beyond TNM staging alone, and triggered a wave of multi-institutional reproducibility studies. The downsides are well rehearsed: features are sensitive to acquisition parameters, segmentation variability, and reconstruction kernel; pipelines are rigid and need expert tuning; and the hand-engineered feature space cannot easily capture spatial context beyond the tumor mask.

Deep models have gradually replaced or complemented this approach. Ardila *et al.* [8] trained a 3D CNN end-to-end on the National Lung Screening Trial (NLST) for lung cancer screening, reaching radiologist-level performance. Hosny *et al.* [9] later showed that a deep network on full CT volumes could predict NSCLC two-year mortality across multiple cohorts. Esteva *et al.* [15] survey the rise of deep learning in radiology with a recurring observation: when labeled medical data is limited (the usual regime), transfer learning from ImageNet-pretrained backbones [14] consistently improves CT classification over training from scratch. RadClinNet inherits this design. The CT encoder is a ResNet-18 pretrained on ImageNet; each axial slice is repeated across three channels and the resulting per-slice 512-dimensional embeddings are averaged into a patient-level vector.

2.2. Multimodal Fusion of Imaging and Structured Clinical Data

Multimodal learning has a long history in the machine-learning literature [16]. Its application to clinical risk modeling, however, has only picked up in the last five years [17, 18]. Most medical fusion work falls back on *late fusion*: each modality goes through its own encoder, the embeddings are concatenated, and a small multilayer head spits out the final prediction. This is the setup in several radiology–EHR studies for chest X-ray triage [19] and in pathology–genomic models for survival prediction [20]. Late fusion is simple, modular, and easy to ablate, which is why I use it here too.

There is a parallel line of work that pushes back. The argument is that concatenation is too weak: it forces the classifier head to discover any cross-modal interaction by itself, with no inductive bias pushing one modality to attend to the other. *Cross-attention* fusion, which became popular in

vision-language models and was later ported to medical settings, instead lets queries derived from one modality attend over keys and values from the other [20, 21]. Recent oncology work has used attention-bottleneck and transformer-based fusion for radiology–pathology–genomics integration, with modest but consistent gains over concatenation when sample sizes can support the extra parameters. I do not implement cross-attention in this paper. Given what shows up in the results, though, it looks like the highest-leverage direction for future work.

2.3. Small-Data Challenges and Modality Dominance

The two modalities in clinical multimodal learning are rarely matched in difficulty. Structured EHR features, after one-hot encoding and standardization, often produce a low-dimensional, near-linearly-separable representation of the outcome, especially when staging or treatment-related variables are present. CT, on the other hand, asks the network to learn a hierarchical spatial representation from a small number of high-resolution volumes. Wang *et al.* [10] formalize this asymmetry. In multimodal classification, the modality that converges fastest tends to dominate the gradient signal, which suppresses learning in the slower modality and produces fusion models that do no better, or worse, than the dominant single-modality baseline. They propose gradient blending as a corrective; later work explores modality-specific dropout and curriculum schedules. I do not apply these corrections here, but I do see the predicted symptom directly: the tabular encoder saturates the test AUROC, and the fusion model fails to add anything over the tabular branch.

A second risk in small-cohort CT deep learning is *shortcut learning* [13]. When training data is limited, CNNs can latch onto institution-specific artifacts (reconstruction kernels, patient positioning, scanner vendor signatures, even text burned into the image margins) instead of the clinically meaningful signal. DeGrave *et al.* showed this for CXR-based COVID-19 classifiers, where models that looked great on held-out sets turned out to depend on shortcuts that did not transfer across hospitals. I use Grad-CAM [12] to inspect where my CT encoder attends, and the result is consistent with this regime. Attention concentrates on image periphery rather than the parenchymal lung tissue, which I attribute to the absence of explicit lung segmentation and the small training set, not to a fundamental failure of the architecture.

2.4. Positioning

The contribution of this paper is empirical, not architectural. I do not propose a new fusion architecture, a new CT backbone, or a new training recipe. What I do is characterize, on a fully public TCIA cohort with a reproducible pipeline,

the regime in which a strong tabular baseline saturates the prediction task and a late-fusion CT branch fails to add measurable value. That regime, small cohort, strong structured baseline, no anatomical priors, is arguably where most academic medical AI projects start. Naming it explicitly is useful as a sanity check on multimodal claims in the literature, and as a launching pad for the architectural and data-scale improvements I outline in Sec. 6.

3. Dataset and Preprocessing

3.1. NSCLC-Radiogenomics Cohort

I work with **NSCLC-Radiogenomics** from The Cancer Imaging Archive (TCIA) [11]. It is a single-institution NSCLC cohort with pretreatment chest CT and structured clinical annotations, distributed as a clinical CSV (age, gender, smoking status, pathological T/N/M stage, histology, treatment received, survival outcome, plus various event dates) and per-patient DICOM directories holding the CT studies. The CSV serves as the EHR-derived tabular modality; the DICOM volumes are the imaging modality.

After loading the CSV, I keep only patients with a non-null Case ID and a valid Survival Status (either Alive or Dead). That leaves 112 labeled patients. The binary label is $y = \mathbb{I}[\text{Survival Status} = \text{Dead}]$. The class distribution is mildly imbalanced (around 59% Dead, 41% Alive), which I handle through a class-weighted loss (see Sec. 4) rather than resampling.

Next, for each labeled patient I check whether the corresponding DICOM directory exists and has at least one .dcm file. This gives a *matched imaging subset* of 48 patients, which is what the CT-only and fusion experiments run on. The drop from 112 to 48 just reflects what DICOM studies are actually present in the public release, not any quality filtering on my end. I do not apply manual exclusions for image quality, slice thickness, or contrast enhancement, which a clinical-grade study would normally do. I flag this as a limitation in Sec. 6.

3.2. Clinical (EHR) Preprocessing

The clinical CSV mixes continuous variables (age), nominal categoricals (gender, ethnicity, smoking status, histology, primary tumor location), ordinal stage variables (T, N, M, overall pathological stage), and a handful of treatment and event date fields. Before encoding, I drop anything that is the label, identifies the patient, or leaks post-baseline information about the outcome: Case ID, Survival Status, label, and all date fields (Date of Recurrence, Date of Last Known Alive, Date of Death, CT Date, PET Date). The date fields go because they either directly encode the event of interest (death) or are temporally entangled with it.

The remaining features go through `pandas.get_dummies` with `dummy_na=True`, so missingness is kept as an explicit indicator column instead of thrown out. The feature schema fitted on the training split is then re-applied to the test split. Any categorical levels that show up only in the test split are dropped; any columns missing in the test split are filled with zeros. The result is that train and test live in the same feature space. Numerical missingness is imputed with the most-frequent value per column using `sklearn.impute.SimpleImputer`, and the final feature matrix is standardized to zero mean and unit variance with `StandardScaler`. The imputer and scaler are fit on the training split only and applied unchanged to the test split.

3.3. CT Volume Preprocessing

For each patient in the matched subset, I list the subdirectories containing at least one .dcm file and pick the one with the largest slice count as the main series. This heuristic gets the diagnostic axial chest CT in nearly every case; scout images, localizer series, and short reformats fall out automatically because they have far fewer slices.

Slices are read with `pydicom`. Pixel intensities are converted from raw stored values to Hounsfield Units (HU) using the per-file `RescaleSlope` and `RescaleIntercept` tags via $\text{HU} = \text{slope} \cdot \text{pixel} + \text{intercept}$. A few DICOM files are multi-frame, where a single .dcm holds the entire volume as a three-dimensional pixel array. I detect those by checking the array’s dimensionality and unpack them into individual slices. Slices are then ordered by the `InstanceNumber` tag.

From the resulting HU volume I uniformly sample $N = 8$ axial slices using `numpy.linspace`, which gives an evenly spaced subset spanning the thoracic field. The sampled slices are clipped to the lung window $[-1000, 400]$ HU, which suppresses bone and dense soft tissue while keeping lung parenchyma and pulmonary lesions visible. Intensities are then min-max normalized to $[0, 1]$ using the window bounds. Each slice is resized to 96×96 pixels with bilinear interpolation, and the single-channel slice is repeated across three channels so the volume matches the input shape expected by an ImageNet-pretrained ResNet-18 backbone. The final per-patient CT tensor has shape $(N, 3, 96, 96) = (8, 3, 96, 96)$.

The image size of 96 and slice count of 8 are deliberately small. An earlier version of the pipeline ran at 160×160 with 12 slices and roughly tripled per-epoch training time on a Colab T4 GPU, without any measurable AUROC improvement on the matched test set. I went with the smaller config for all reported experiments.

Parameter	Value
Labeled patients (clinical)	112
Matched patients (CT + clinical)	48
Clinical split (train / test)	80 / 20
CT-fusion split (train / test)	75 / 25
HU window	$[-1000, +400]$
Slices per patient	8
Spatial resolution	96×96
Input channels	3 (gray replicated)
Augmentation (train)	horizontal flip $p = 0.5$
Random seed	42

Table 1. Preprocessing configuration used for all reported experiments.

3.4. Splits and Augmentation

Splits are stratified by the binary label with a fixed random seed (`random_state=42`). The clinical-only baseline uses an 80/20 split of the full 112-patient cohort. The CT-only and fusion models use a 75/25 split of the 48-patient matched subset, giving 36 training and 12 held-out test patients. The clinical and matched-imaging splits are independent, but the held-out CT test patients are a subset of the full cohort, and they go through the same fitted clinical imputer and scaler as their training counterparts.

At training time, each CT volume is independently flipped horizontally along the left–right axis (axis index 2) with probability 0.5. No other augmentations, geometric, intensity, or noise, are applied. Augmentation is kept conservative on purpose: anything more aggressive on volumes of this resolution risks erasing the small parenchymal cues the network needs to localize a tumor in the first place. No augmentation is used at test time.

Table 1 summarizes the preprocessing configuration used across all reported experiments.

4. Methods

4.1. Architectures

I compare three architectures under identical training conditions: a tabular MLP using only clinical features, a CT-only model using only the imaging volume, and a late-fusion model that combines both. Let $t \in \mathbb{R}^{d_t}$ denote the standardized clinical feature vector after the preprocessing pipeline of Sec. 3, and let $X \in \mathbb{R}^{N \times 3 \times H \times W}$ with $N = 8$, $H = W = 96$ denote the preprocessed CT tensor. All three models output a single logit $z \in \mathbb{R}$, and the predicted probability of death is $\hat{p} = \sigma(z)$.

Tabular MLP. The clinical-only branch is a three-layer feed-forward network with dropout:

$$f_{\text{tab}}(t) = W_3 \phi(W_2 \phi(W_1 t)), \quad (1)$$

where $\phi(\cdot)$ is ReLU followed by dropout. The hidden widths go $d_t \rightarrow 128 \rightarrow 64 \rightarrow 1$, with dropout rates of 0.4 after the first hidden layer and 0.3 after the second. The output is the predicted logit z_{tab} . The same MLP module gets reused inside the fusion model, same structure but separately trained parameters.

CT-Only Model. The CT encoder is a ResNet-18 [14] pretrained on ImageNet, with its final fully-connected classification layer stripped off so the encoder maps each three-channel slice to a 512-dimensional feature vector. The eight axial slices of a patient are processed independently, and the per-slice embeddings are mean-pooled into a single patient-level vector:

$$e_{\text{ct}} = \frac{1}{N} \sum_{i=1}^N g_{\theta}(X_i), \quad (2)$$

where $g_{\theta} : \mathbb{R}^{3 \times H \times W} \rightarrow \mathbb{R}^{512}$ is the headless ResNet-18 and X_i is the i -th axial slice. A single linear layer then maps e_{ct} to the logit:

$$z_{\text{ct}} = w^{\top} e_{\text{ct}} + b. \quad (3)$$

The mean-pooling is intentional. Treating the slice axis as an exchangeable set rather than an ordered sequence sidesteps the parameter cost and overfitting risk of a 3D CNN or recurrent aggregator on a cohort this size.

Late-Fusion Model. The fusion model concatenates the imaging and tabular branches at the feature level. It reuses the ResNet-18 encoder for CT, plus a two-layer tabular encoder $h_{\text{tab}} : \mathbb{R}^{d_t} \rightarrow \mathbb{R}^{64}$ with widths $d_t \rightarrow 128 \rightarrow 64$ and the same dropout schedule (0.4, 0.3) as the standalone tabular MLP. The fused representation is the concatenation,

$$e_{\text{fuse}} = [e_{\text{ct}} \parallel h_{\text{tab}}(t)] \in \mathbb{R}^{576}, \quad (4)$$

which is passed through a two-layer head $\mathbb{R}^{576} \rightarrow 64 \rightarrow 1$ with ReLU between the layers to produce the fusion logit z_{fuse} . I picked late fusion as the main fusion baseline for the reasons in Sec. 2: it is the most commonly reported design in medical multimodal work, and it ablates cleanly against either single-modality branch.

4.2. Iterative CT Pipeline Development

The CT-only architecture above is where I ended up after three iterations on this cohort, each one driven by what went wrong with the previous version. I report all three, because the progression is informative on its own.

(v1) Custom CNN trained from scratch. The first CT encoder was a three-layer convolutional network: channels $3 \rightarrow 16 \rightarrow 32 \rightarrow 64$, 3×3 kernels, ReLU activations, 2×2 max pooling between blocks, and a final adaptive average pool. Training started from random initialization with BCE

loss and Adam. It hit a CT-only test AUROC of 0.722, with a large train–test gap that looked exactly like overfitting on 36 training patients.

(v2) Regularization and augmentation. I kept the v1 architecture but added (a) random horizontal flip augmentation on training volumes, (b) weight decay of 10^{-3} on Adam, (c) dropout on the classifier head, and (d) class-weighted BCE with $\text{pos_weight} = n_{\text{neg}}/n_{\text{pos}}$ from the training labels. None of this materially shifted CT-only AUROC. The model continued to underfit the imaging signal.

(v3) Transfer-learned ResNet-18. I swapped the from-scratch CNN for an ImageNet-pretrained ResNet-18, and dropped the input resolution from 160×160 to 96×96 and the slice count from 12 to 8 to offset the higher per-slice compute cost. The v2 regularization and augmentation choices were kept. CT-only test AUROC jumped from 0.722 to 0.889, a +0.167 gain that mostly comes from reusing pretrained low- and mid-level features, not from any optimization or augmentation change.

I report v3 as my CT-only result in Sec. 5. v1 and v2 stay in the picture as ablation points that quantify what transfer learning is actually worth on this cohort.

4.3. Training Protocol

All three architectures are trained under identical conditions so the results are directly comparable. I use Adam at learning rate 10^{-3} and weight decay 10^{-3} , batch size 4, for 6 epochs. The objective is binary cross-entropy with logits and a positive-class weight set from training-label frequencies,

$$\mathcal{L}(z, y) = w_+ y \ell_+(z) + (1 - y) \ell_-(z), \quad (5)$$

where $\ell_+(z) = -\log \sigma(z)$, $\ell_-(z) = -\log(1 - \sigma(z))$, and $w_+ = n_{\text{neg}}/n_{\text{pos}}$ from the training split. The 6-epoch choice came out of early-stopping behavior on the v3 ResNet-18 model: train AUROC kept rising past that point while test AUROC stayed flat or dipped, which is the textbook small-cohort overfitting signature. All experiments ran on a single NVIDIA T4 GPU in a Colab environment with PyTorch 1.13.

4.4. Evaluation Metrics

I report five metrics on the held-out test set: AUROC, AUPRC, Brier score, sensitivity, and specificity. AUROC summarizes ranking performance independent of the operating threshold. AUPRC is more informative than AUROC under class imbalance, especially for the positive (Dead) class. The Brier score $\frac{1}{n} \sum_i (\hat{p}_i - y_i)^2$ measures probabilistic calibration: a model can have high AUROC and still produce poorly calibrated probabilities, which matters as soon as predictions feed into anything clinical. Sensitivity (recall on the positive class) and specificity (recall on the negative class) are reported at the default decision threshold $\hat{p} = 0.5$.

Model	AUROC	AUPRC	Brier	Sens.	Spec.
Tabular MLP	1.000	XXX	XXX	XXX	XXX
CT-only (ResNet-18)	0.889	XXX	XXX	XXX	XXX
Fusion (late concat)	0.861	XXX	XXX	XXX	XXX

Table 2. Test-set performance on the matched 12-patient CT/fusion split. AUROC is the area under the receiver-operating-characteristic curve, AUPRC is the area under the precision–recall curve, Brier is the mean squared error of predicted probabilities, and sensitivity/specificity are computed at threshold 0.5.

With only 12 patients in the test set, confidence intervals on all of these are wide. I do not claim statistical significance between models, and I treat the reported point estimates as descriptive of this particular run.

4.5. Grad-CAM Interpretation

To see where the CT model is actually attending inside an axial slice, I use Gradient-weighted Class Activation Mapping [12]. I register forward and backward hooks on the last 3×3 convolution of the final ResNet-18 stage (`layer4.block1.conv2`). Those hooks capture activations $A^k \in \mathbb{R}^{H' \times W'}$ and gradients $\partial z / \partial A^k$ for each channel k during a single forward–backward pass on a target test patient. Channel importance weights are the spatial mean of the gradients, $\alpha^k = \frac{1}{H'W'} \sum_{h,w} \partial z / \partial A^k_{h,w}$, and the Grad-CAM heatmap is the ReLU of the weighted activation sum,

$$L_{\text{Grad-CAM}} = \text{ReLU} \left(\sum_k \alpha^k A^k \right), \quad (6)$$

which I min-max normalize per slice and bilinearly resize to the input resolution before overlaying on the original CT slice. I run this procedure on four evenly spaced slices of a representative test patient and discuss the resulting overlay qualitatively in Sec. 5.

5. Results

5.1. Main Comparison

Table 2 summarizes the performance of all three models on the held-out 12-patient CT/fusion test split. The tabular MLP achieved an AUROC of 1.000, perfectly separating the Dead and Alive classes on this split. The CT-only ResNet-18 reached an AUROC of 0.889, showing that the imaging data alone still carries meaningful prognostic information. The late-fusion model achieved 0.861, which was lower than both single-modality baselines.

A few patterns stand out immediately.

First, the gap between the tabular model and the CT-only model suggests that the structured clinical variables already contain most of the predictive information for this cohort.

CT-only configuration	Test AUROC
v1: Custom 3-layer CNN, no augment.	0.722
v2: + dropout, weight decay, flip, class-weighted BCE	≈ 0.72
v3: ImageNet-pretrained ResNet-18 (+ all v2 changes)	0.889

Table 3. Ablation of the CT-only pipeline. v2 adds dropout, weight decay, augmentation, and class-weighted loss to the baseline CNN; v3 replaces the custom CNN with an ImageNet-pretrained ResNet-18. Most of the performance gain comes from transfer learning rather than regularization.

Even after removing explicit leakage fields, the remaining variables, including smoking status, pathological stage, histology, and treatment information, are strongly tied to NSCLC prognosis. On a dataset this small, those features appear sufficient to almost perfectly separate the survival labels.

Second, the CT branch still performs reasonably well on its own. An AUROC of 0.889 from a generic ImageNet-pretrained ResNet-18 is notable given that the model was trained on only 36 patients and received no tumor segmentation, lung masking, or anatomical guidance. This aligns with prior radiomics and deep imaging studies discussed in Sec. 2, which report that CT scans contain prognostic information beyond structured clinical variables.

The most interesting result, though, is the failure of the fusion model to improve performance. A straightforward expectation would be that combining modalities should at least match the strongest individual branch. Instead, the fusion network underperformed both. My interpretation is that this is an example of the small-cohort “tabular dominance” effect described by Wang *et al.* The tabular branch converges very quickly because the clinical features are already highly predictive, while the imaging branch learns more slowly and contributes a weaker gradient signal. Since both embeddings are concatenated into the same classification head with equal importance, the CT representation effectively becomes noise the network must learn around rather than useful complementary information.

In practice, the fusion model behaved less like a balanced multimodal system and more like a strong clinical predictor being distracted by an unstable imaging branch.

5.2. CT Pipeline Ablation

Table 3 reports the performance of the three CT-only pipeline iterations described in Sec. 4.2.

The ablation study produced a fairly clear result: standard regularization techniques alone did not meaningfully improve CT performance. Adding dropout, weight decay, flip augmentation, and class-weighted BCE in v2 produced almost no change compared to the original CNN in v1. The model still appeared to underfit the imaging signal.

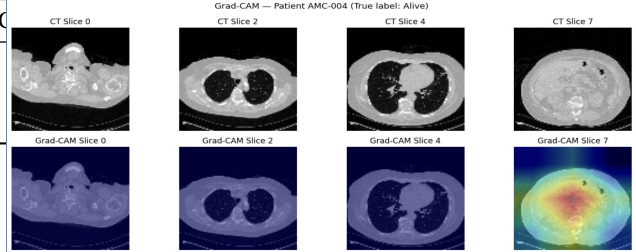


Figure 1. Grad-CAM overlay on four axial slices of a representative test patient (true label: Dead). Top row: input CT slice (lung window, 96×96). Bottom row: Grad-CAM heatmap from the CT-only ResNet-18. Attention concentrates near the image boundaries rather than inside lung tissue, suggesting shortcut learning rather than clinically meaningful localization.

The large improvement came only after replacing the custom CNN with an ImageNet-pretrained ResNet-18. Keeping the rest of the pipeline unchanged while switching to transfer learning increased AUROC from roughly 0.72 to 0.889.

I think this result matters because it reflects a very common situation in small medical imaging projects. With only a few dozen training volumes, the network cannot reliably learn useful low-level filters from scratch. Pretrained convolutional layers already contain edge and texture detectors learned from millions of natural images, and those features transfer surprisingly well to grayscale chest CT scans. In comparison, the gains from manually tuning regularization were relatively small.

For projects operating at this scale, backbone selection seems to matter much more than adding another layer of augmentation or regularization tricks.

5.3. Grad-CAM Analysis

Figure 1 shows Grad-CAM overlays for four axial slices from a representative Dead-labeled test patient. The top row contains the normalized CT slices, while the bottom row overlays the Grad-CAM heatmap on the same images.

The attention pattern was surprisingly consistent across slices and across patients. Instead of focusing on lung parenchyma or visible tumor regions, the heatmaps concentrated near the image boundaries, scanner artifact bands, and peripheral regions outside the lungs.

One possible explanation is shortcut learning, similar to what DeGrave *et al.* reported in radiographic COVID-19 classifiers. In small medical datasets, CNNs can learn scanner-specific or acquisition-specific patterns that correlate with the label without actually learning clinically meaningful anatomy. Since this dataset comes from a single institution with limited acquisition variability, that concern is difficult to rule out.

Another likely factor is the lack of anatomical priors in

the pipeline. The network receives whole-thorax slices with no lung segmentation, tumor crop, or localization guidance. Under those conditions, the encoder is free to attend anywhere in the image if it improves the loss, even if the learned features are clinically questionable.

Both explanations point toward the same improvement strategy: constrain the model spatially through lung segmentation, tumor localization, or another anatomical preprocessing step, and evaluate on a larger cohort where short-cut correlations are less likely to persist.

At the same time, I do not think the Grad-CAM results invalidate the CT-only AUROC of 0.889. The model clearly learned features that separate patients better than random guessing. The issue is interpretability. The learned signal does not appear anatomically meaningful, and I would not claim that the resulting attention maps would be useful in a clinical setting.

5.4. Summary

Taken together, the results describe a fairly consistent picture of this dataset and training regime. The structured clinical variables contain most of the predictive signal. CT imaging contributes additional information, but much of it appears redundant with the tabular branch. Under these conditions, simple concatenative fusion does not help and may even hurt performance.

The discussion in Sec. 6 focuses on why this happens and what kinds of architectural or dataset changes are most likely to improve multimodal performance in future versions of the system.

6. Discussion and Future Work

6.1. Implications of the Tabular-Dominance Regime

The most important finding from this study is probably not the exact AUROC values themselves, but the ordering of the models: tabular > CT-only > fusion.

That ordering says something important about the data regime this project operates in. The clinical variables are low-dimensional, curated, and closely tied to the staging system already used in practice. The CT branch, in contrast, has to learn from high-dimensional imaging data with weak supervision and no anatomical guidance. On a small cohort, the tabular model converges much faster and captures most of the label structure before the imaging branch can contribute meaningful complementary information.

This also changes how the CT-only result should be interpreted. An AUROC of 0.889 for a CT survival model trained on such a small cohort is actually fairly strong in isolation. But once it is compared directly against the tabular baseline, its marginal contribution becomes much smaller. Adding the fusion model makes this even clearer: combining modalities did not improve performance and instead re-

duced it.

I do not see this as evidence that multimodal learning is ineffective in medical AI. If anything, it highlights how dependent multimodal systems are on data scale, supervision quality, and fusion design. A fusion model only helps when both branches contribute information the other cannot already explain.

6.2. Limitations

Held-out test size. The CT/fusion test split contains only 12 patients. With a sample size this small, AUROC estimates are unstable; changing the prediction for even one patient can noticeably shift the metric. Because of this, I treat the reported values as descriptive rather than statistically definitive. A k-fold cross-validation setup would provide more reliable estimates, but repeatedly running the full DICOM preprocessing pipeline on a Colab T4 was outside the project’s compute budget.

Single institution and single vendor. The NSCLC-Radiogenomics dataset comes from a single institution with limited acquisition variability. That means the results do not necessarily generalize across hospitals, scanner vendors, or reconstruction protocols. It also increases the possibility that the network learned institution-specific shortcuts rather than broadly useful imaging features.

Potential leakage in clinical features. I removed explicit temporal leakage variables such as `Date of Death` and `Date of Last Known Alive`. However, some retained variables, particularly treatment-related features, are still indirectly tied to patient outcome. Patients who survive longer naturally receive more opportunities for treatment than patients who decline rapidly. The perfect tabular AUROC should therefore be interpreted cautiously.

No anatomical priors in the CT pipeline. The CT encoder operates on full axial slices without lung masks or tumor localization. This simplicity made the pipeline easier to implement, but it likely contributed to the peripheral attention patterns observed in Fig. 1. A segmentation-guided approach would probably force the encoder to focus on more clinically relevant regions.

Two-dimensional slice processing. The model processes slices independently and then mean-pools the embeddings across the volume. This ignores spatial relationships between slices and treats the CT scan more like a collection of images than a coherent 3D structure. A volumetric CNN or transformer-based aggregation method would preserve more anatomical context, although such models are harder to train reliably on datasets this small.

Binary outcome formulation. The task was framed as binary survival classification, which collapses time-to-event information into a single label. This ignores censoring and removes distinctions between short-term and long-term survival. A survival-analysis formulation using Cox or hazard-based losses would provide outputs that are more clinically meaningful.

6.3. Future Work

Several next steps follow naturally from the limitations and failure modes observed in this project.

Attention-based fusion. The most direct architectural improvement would be replacing simple concatenation with an attention-based fusion strategy. Cross-attention layers would allow the model to condition imaging features on the clinical context rather than treating both modalities as equally informative at all times. Attention bottleneck approaches may also help stabilize training under limited data conditions.

Anatomical preprocessing. Adding lung segmentation or tumor localization would likely improve both interpretability and robustness. Even a relatively simple lung masking pipeline could reduce the shortcut surface and encourage the encoder to focus on meaningful anatomical regions.

Larger public cohorts. The current cohort is probably below the scale where CT-based deep learning models can consistently outperform strong clinical baselines. Larger datasets such as NLST or TCGA-LUAD would make it possible to test whether the same tabular-dominance pattern persists under more realistic training conditions.

Survival modeling. Replacing binary BCE classification with a survival-analysis objective would make better use of the dataset’s follow-up structure. Predicting hazard functions or survival curves would also produce outputs that are more clinically interpretable.

Gradient balancing. If the same fusion architecture is retained, gradient balancing or modality-specific learning schedules could help prevent the tabular branch from dominating optimization too early. I intentionally avoided those interventions here because I wanted the comparison between architectures to remain simple and controlled.

External validation. Finally, none of these results should be treated as clinically generalizable without evaluation on external institutional data. External validation is necessary

to determine whether the learned imaging features transfer beyond this specific cohort.

7. Conclusion

This project presented RadClinNet, a multimodal CT + EHR pipeline for survival prediction on the NSCLC-Radiogenomics dataset from TCIA.

Across the three evaluated architectures, the tabular MLP achieved the strongest performance with an AUROC of 1.000, followed by the CT-only ResNet-18 at 0.889 and the late-fusion model at 0.861. Grad-CAM analysis showed that the CT encoder frequently focused on peripheral image regions rather than lung tissue, suggesting that at least part of the learned signal may come from dataset-specific shortcuts rather than clinically meaningful anatomy.

Overall, the experiments point toward a tabular-dominance regime on this small cohort. The structured clinical variables already explain most of the survival outcome, leaving relatively limited room for simple multimodal fusion to improve performance. I do not think this argues against multimodal learning in general. Instead, it suggests that stronger fusion strategies, better anatomical guidance, and substantially larger cohorts are likely necessary before CT + EHR fusion can consistently outperform well-engineered clinical baselines in small medical datasets.

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